

ON TRANSCENDENTAL SYNTAX: A KANTIAN PROGRAM FOR LOGIC?

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Abstract

We present the general lines of the transcendental syntax program, recently proposed by J.-Y. Girard as a natural development of research in the field of linear logic. Building upon a “geometrical” interpretation of the so-called *Curry-Howard correspondence* between logical proofs and typed λ -terms, this program aims at finding an internal explanation of logical rules.

The Kantian inspiration of the program resides in the look for inner explanations, as opposed to semantical ones, and in the need for a synthetic explanation of the conditions of possibility of the use of formal languages in logic.

Such a questioning of the syntactic dimension of mathematical logic, as opposed to the analytic “dogma” of the philosophical primacy of linguistic notions, so as the criticism of the syntax/semantics distinction, are indeed two of the main theoretical challenges issued by the developments of proof-theory, and to which the newborn program of transcendental syntax aspires to provide both a technical and philosophical answer.

1 *From linear logic to transcendental syntax*

At its birth in 1987 (see Girard (1987)), linear logic obeyed all the standards a good logical theory had to: sequent calculus worked especially well and a promising graph-theoretical reformulation of it, *proof-nets*, played a synthetic role analogous to the one played by natural deduction with respect to intuitionistic sequents; syntax could moreover be proven to be sound and complete with respect to *phase semantics*, which was considered then as the “model-theory”, so to say, of linear logic.

The application to logic of the algebraic notion of *linearity*, which provided from the start a powerful bridge with linear algebra (exposed by the development of *geometry of interaction*), had been the result of an investigation by Jean-Yves Girard (the founder of linear logic and inspirer of almost all of its developments) on the mathematical properties of typed λ -calculi. Linear logic is indeed one of the children of the prolific encounter, made official under the name of *Curry-Howard correspondence*, between proof-theory “à la Gentzen”, centered on the *Hauptsatz* (the cut-elimination theorem), and the functional paradigm in theoretical computer science, focused indeed on λ -calculus.

Research on typed λ -calculi usually privileges, indeed, an approach directed to the *dynamical* features of logic: proofs, as isomorphic with programs, are taken as mathematical objects that must not only be *constructed* following the rules, but also be *executed* following the dynamics of those rules.

The duality construction/execution is made explicit by the so-called *formulae-as-types* correspondence, by which logical formulas are associated with types, that is, with sets of proofs satisfying the norms associated by logical rules to those formulas; this interpreta-

tion can be read in two complementary ways: *à la Church*, proof-terms are built inductively following logical rules, and are *a priori* typed, while *à la Curry* proof-terms are pure λ -terms, and typing is given *a posteriori* by norms which assure termination. For instance, a (normal) proof-term of $A \Rightarrow B$ *à la Church* has the form $\lambda x^A.t$ and is built following the rule

$$\frac{x : A \vdash t : B}{\vdash \lambda x^A.t : A \Rightarrow B} \quad (1)$$

which isn't but a proper reformulation of the introduction rule for implication. On the other hand, its corresponding proof-term *à la Curry* is just the subjacent pure term $\lambda x.t$, provided that, for every proof-term u of type A , the computation $(\lambda x.t)u$ terminates and gives a proof-term of type B (you may note that typing *à la Church* follows the introduction rules, whereas typing *à la Curry* follows the elimination rules, so that the equivalence of the two formulation - as far as one stays at the first order - corresponds to the symmetry between introduction and elimination rules, that is, to a cut-elimination theorem).

Logical syntax can thus be seen both as a constructive tool, enabling the formation of (normalizing) typed terms, and as a constraining one, imposing a *custom* on pure terms (and their socialization) in order to force termination. As a remarkable example, we mention the impossibility of typing the diverging, “incestuous”, λ -term $(\lambda x.(x)x)\lambda x.(x)x$, which indefinitely reproduces itself.

Through linear logic, this revised, “normative”, version of constructivism was interpreted in a surprising geometrical way: in proof-net theory, typed λ -terms are represented *à la Church* as graphs, whereas in geometry of interaction they are represented *à la Curry* as operators (in the sense of operator algebras).

Nevertheless, it was only later, more or less at the turn of the century, that it was

explicitly realized that adopting such tools involved something more than just interesting technical contributions, since the mathematics of proof-nets seemed not to fit well in the traditional syntax/semantics frame: for instance, since the geometrical character of the graphical interpretation of a λ -term lies in parallelism, typing constraints are given through an (essentially topological) criterion which assures that a graph corresponding to a well-typed term can be sequentialized. On the contrary, incorrect (i.e. not well-typable) nets do not admit of any sequentialization, since it happens that any attempt is blocked by the existence of *cycles* (i.e. closed paths) in the net.

Now, the *tests* that a net of type A must pass in order to satisfy the sequentiality criterion induce a proof of its negation $\sim A$ in a sequent calculus obtained from standard sequent calculus by adding the so-called *Daimon*-rule, which allows to immediately derive every sequent:

$$\overline{\vdash \Gamma} \quad \text{✕} \tag{2}$$

Proofs with *Daimons* are not standard proofs, because everything can be derived with them, but they make perfect sense as sequential objects: they behave very well as tests, that is as what a correct proof should refute (for instance, they preserve cut-elimination). The interaction between the proof-net and its tests can thus be seen as a sort of cut-elimination between an alleged proof of A and an alleged proof of its negation, in order words, as a *dispute*. As a consequence, it appears that a net is actually typable (i.e. sequentializable) if and only if, when interacting in disputes, it doesn't run into vicious circles, preventing sequentialization.

This replaces the familiar duality proofs/models (i.e. proofs/refutations) by a duality proofs/counterproofs, which is much more satisfactory. Girard (1988)

It is of great importance to remark that sequentialization, by stating that a net which passes all tests, that is, that refutes all refutation, can be built following the rules of sequent calculus, can be interpreted as a *completeness theorem* of a radically “internal” nature: usual “external” completeness is indeed formulated with respects to outer counter-models, whose role is here played by tests, i.e. alleged proofs of the dual formula.

In this sense, proof-net theory can be seen as an elegant synthesis of the “internalist” perspective on logic which has been gradually emerging in the research in proof-theory following Gentzen’s techniques. It is indeed a well known result in proof-theory, whose foundational relevance we believe has not yet been highlighted enough in the literature, that Gentzen’s *Hauptsatz* has, as corollaries, purely *internal* versions of the soundness and the completeness theorems for first-order logic, which establish a sort of duality similar to the one just sketched (see Pistone (2011) for an extended discussion), given by the fact that a formula has a proof if and only if it admits of no countermodels. However, if we accept the *a priori* separation of syntax and semantics, there is no way to improve this duality, that is, to let proofs and models actually interact, as is the case for proof-nets.

As a consequence of these remarks, the mathematical study of cut-elimination procedures seems of crucial importance even for the investigation of the foundational aspects of logic: for instance, can completeness and soundness, rather than semantical (i.e., external) verdicts, result as products of the internal symmetries of logical syntaxes, so that a serious mathematical distinction between arbitrary (purely conventional) rules and logically correct ones can be drawn?

The road actually taken by Jean-Yves Girard is indeed that of looking for a radically new paradigm for logic, investigating logical formalization with the aid of genuinely

mathematical tools (as linear algebra, topology and functional analysis): this is what constitutes the central point of the program of *transcendental syntax*, launched by him in 2011 through a series of articles and lectures (Girard (2011a,b, 2012))

If we have a look at the works just mentioned, we notice that Girard's mathematical research is more and more inspired by philosophical intuitions, to the point that his project made explicit reference to Kant even in its name.

Mais, dans les limites de sa compétence, celle d'un catadioptré qui réfléchit la technique, l'intervention de la philosophie me semble inévitable: c'est bien la seule nécessité que je vois. Girard (2011b)

In the section that follows we try to give a hint at the aporias that the traditional "format" of logic seems to raise and that the *transcendental syntax* project aspires to resolve. In analogy with Kant's transcendentalism, indeed, the program tries to raise the issue to *deduct*, that is, to legitimize the authority exercised by logical languages: as Girard puts it, the point would be to pose in logic "the more general question «Is this appropriate?», a *normative* query which encompasses the evaluative questioning about truth" (Girard (2011a)). In section 3 we try then to show in what sense transcendental syntax can aspire to provide a technical ground for a philosophical reconsideration of those aporias.

As already pointed out, many of the themes here proposed have been and are still highlighted and discussed in other research programs in mathematical and philosophical logic (for instance in the proof-theoretic semantics programs by, among others, Martin-Löf, Prawitz and Dummett). On the other hand researchers within such programs have, so far, never engaged a serious confrontation with results coming from linear logic. It seems to us that an interplay between those well-established philosophical traditions and the

actually rising perspective of transcendental syntax (which, as every newborn perspective, is still full of gaps) would be of great interest. For this reason, rather than just sketching some similarities and some (relevant) differences, we have preferred to leave a serious and rigorous confrontation with those perspectives as the topic of a future paper.

2 *Some limits of logical analysis*

Two are, in our opinion, the aspects which characterize the paradigm within which mathematical logic is usually described (for instance, in manual books), and which are put into question by the transcendental syntax program: firstly, the necessity to provide, in advance to the definition of “a logic”, a formal description of the syntactic devices to be used within such a definition (i.e. a definition of what is to be considered a *formal language* and a *formal system*); secondly, the fact that the rules and axioms which constitute, within a formal system, “the logic” under consideration, have to be validated by reference to an *interpretation*, that is, a function assigning specific values, belonging to a certain mathematical structure, to linguistic entities, so that the interpretations of those rules and axioms result in transformations preserving some of those semantic values (typically, truth).

These aspects reveal indeed a conception of language and logic in which the presentation of a rigorous formal language is a precondition for the formulation of logical notions and in which logic itself provides a connection between syntax and an external evaluative dimension (semantics). In paragraph 2.1 we propose a brief recognition of some of those historical views on foundation whose influence has been, as it is widely accepted in the literature, decisive for the entrenchment of such a conception. In paragraphs 2.2 and 2.3

we propose two arguments against it, elaborated on the basis of ideas coming from transcendental syntax: a first one based on Kant's conception of transcendental philosophy and his distinction between analysis and synthesis, a second one focused on the difficulty to escape some forms of circular justification of rules.

2.1 *The linguistic turn and logic: two key examples*

The demand for rigour has characterized a large component of the philosophical debate of last century and was at basis of the birth of mathematical logic, originally intended as the fundamental tool for a reform, in the name of mathematical rigor, of the logic inherited from the tradition (dating back to Aristotle), and at the same time as independent means to get to grips with the heavy transformations mathematics was passing through, by providing a solid ground for them.

As it is widely known, indeed, the nineteenth century was, for mathematics (especially analysis), a period of great and profound renewal, characterized by the process of the so-called *arithmetization of analysis*, achieved through a progressive reconciliation of methods and contents coming from differential and integral calculus, characterized by the frequent appeal to non-formal tools (as spatial intuition), with those coming from arithmetics and basic algebra.

The birth of modern mathematical logic, in which the work of Gottlob Frege played a decisive role, was tied to the increasing exigency of rigour, due to the inaccuracy of the language adopted by mathematicians:

To prevent anything intuitive from penetrating here unnoticed, I had to bend every effort to keep the chain of inferences free of gaps. In attempting to comply with this

requirement in the strictest possible way I found the inadequacy of language to be an obstacle; no matter how unwieldy the expressions I was ready to accept, I was less and less able, as the relations became more and more complex, to attain the precision that my purpose required. This deficiency led me to the idea of the present ideography.

Frege (1972)

Frege's grammatical concern, here clearly expressed, was historically crucial for the development of that system of philosophical, linguistic and logical ideas which is usually referred to as the *linguistic turn*, and which implies that a proper grammatical *analysis* of the linguistic expressions adopted to present a given philosophical position has to be constitutive of its philosophical evaluation. Accurate linguistic formalizations, as prior to the elaboration of specifically philosophical theses, started to constitute a diffuse methodology mainly through the work of Frege himself (think of his philosophical discussion on the notion of number in Frege (1950)) and the influence of Russell (as a compelling example, consider Russell (1905)).

The ideal of Frege and his followers was in fact that of a complete explication, through a rigorous syntactical analysis, of logical inferences, as basilar to a proper justification of mathematical proofs and philosophical argumentations. Such an ideal was accomplished through the introduction of formal systems (of which Frege's *ideography* constitutes one of the first examples): in a formal system, all that is required to evaluate a syntactical expression is its *analysis*, that is, the decomposition of it obtained traveling back its explicit generative path along the rules of the system, reaching for a complete decomposition up to the axioms in a finite amount of time. Rules and axioms assume thus the role of atoms whose disputability is relegated out of the domain of rigorous philosophical

disputations. Clear symptoms of this analytical stand can be found in the Wittgensteinian *Tractatus*, where it is clearly stated that questions about grammatical choices are to be considered as devoid of sense, or in Carnap's conventionalism, see Carnap (1950), where those questions are styled *external* and destined to merely "pragmatical" considerations.

Though Frege's ideas were probably the most influential for analytic philosophy, relevant work in the foundations of mathematics, at the turn of the nineteenth and twentieth century, was made following different ideas. It seems quite hard, for example, to conceive the works on arithmetics by Kroenecker, on transfinite numbers by Cantor, not to speak of the intuitionistic reconstruction of analysis carried over by Brouwer, as inspired by an essentially grammatical questioning.

Hilbert's position, in this debate, seems especially significant to us, in that his foundational work retains the grammatical concerns of the "linguistic turn" but, as shown by the quotation below, does not present syntactical rigour as a pre-condition for the objectivity of mathematics, but rather as a means to achieve mathematical deepening, necessary for a more satisfactory justification of its foundations: the work on foundations is not then required to give objectivity to a discipline which has not it, but to achieve a more profound understanding of its contents and of its reliability:

The procedure of the axiomatic method, as it is expressed here, amounts to a deepening of the foundations of the individual domains of knowledge - a deepening that is necessary for every edifice that one wishes to expand and to build higher while preserving its stability. Hilbert (1996a)

The grammatical concern seems thus exclusively functional to the aim of providing mathematical proofs of consistency, in accordance with the identification, typical

of Hilbert's position, of the absence of contradictions in an axiomatic system with the existence of the mathematical objects referred to by that system.

If contradictory attributes be assigned to a concept, I say, that mathematically the concept does not exist. So, for example, a real number whose square is -1 does not exist mathematically. But if it can be proved that the attributes assigned to the concept can never lead to a contradiction by the application of a finite number of logical inferences, I say that the mathematical existence of the concept (for example, of a number or a function which satisfies certain conditions) is thereby proved. Hilbert (1996b)

Frege's and Hilbert's views on the foundations of mathematics diverge on an important aspect which regards the role of language: the rigorous (in the sense of admitting a serious metamathematical enquiry) grammatical presentation is in fact for the latter not a precondition, but a significant part, if it eventually leads to a proof of non-contradiction, of a process of in-depth understanding and securing of mathematics. Syntactical refinement then, following this way of interpreting Hilbert, would be part of the growth of mathematical thought itself. Meta-mathematical justifications of formal systems, if ever possible (what has been definitely shown not to be the case), would have indeed played the role of a mathematical "supreme court" over formal systems, in contrast with Wittgenstein's "whereof one cannot speak, thereof one must be silent" (Wittgenstein (2001)).

Thus, even if under a quite different understanding of their role in foundations, the focus on formal systems constituted a central tenet of both Frege's and Hilbert's programs, and, through these, it stood out as the most successful response to the quest for rigor in mathematics, with respect to other solutions, as the Brouwerian one, centered on intuition.

2.2 *Analysis and the transcendental subject*

Kant's transcendental philosophy, with its vocabulary widely borrowed from the language of lawyers, hinges on the thesis that the objectivity which pertains to mathematical and empirical truths (and even moral or political duty) has a genuinely *normative* character: rather than describing an abstract metaphysical reality, the indisputable truths provided by Euclidean geometry or Newtonian physics constitute something which *ought* to be held as such, in virtue of the "legal" principles which sustain them and which are valid *a priori* for every one, since constitutive of subjectivity in general: Kant's celebrated "Copernican revolution" consisted indeed in the idea that the transcendental subject, as determined by the system of those principles, plays a "legislative" role, by imposing on nature the forms necessary for it to be experienced.

The aim of Kantian philosophy was thus to expose those constitutive principles and, when possible, to *deduct* them (in the juridical, not logical sense), that is, to legitimize the power they actually exercise on experience. In this sense, typical of transcendentalism is the refusal of every transcendent form of justification, as resting on theological, ontological or psychological arguments: the verdict as to the legitimacy of the use of a given system of *a priori* concepts is entirely under the jurisdiction of the "tribunal of reason", that is, it has to be the result of a purely *internal* enquiry.

If this was the case for the immanent justification of scientific knowledge (through the principles of transcendental esthetics and analytics) and moral law (through the categorical imperative), so as for the rejection of the ambitions of metaphysics (through transcendental dialectics), similar considerations can be made concerning Kant's concep-

tion of logic¹: the laws of logic must indeed be expressed in terms of a logical syntax solidly grounded in *a priori* principles of the understanding, otherwise knowledge of logical truths would be indistinguishable from the empirical knowledge of a given historical grammar:

Die Grammatik ist [...] nur eine Disciplin, die Logik eine Wissenschaft; Doktrin.

Die Unterschied zwischen Disciplin und Wissenschaft oder Doctrin ist der; Bei der ersteren kann man nicht wissen, warum etwas so und nicht anders ist, bei der andern aber hat man einen inneren Beweiss davon. Kant (1966) (quoted in Capozzi (2006))

Grammar is just a discipline, logic a science; doctrine. The difference between discipline and science or doctrine is that in the former you may not know why something is so and not otherwise, in the latter you can know this by an internal demonstration
[our translation]

Since logic, dating back to the times of Aristotle, is essentially *analytic*, a transcendental approach to logic has to look, far from any appeal to ontology or psychology, for the constitutive principles of the analysis of what he calls the “judgements of experience”, in order to find means, if any, for an inner justification of the application of logical rules to experience; Kant achieves this by focusing on the forms of *synthesis*² of representations, that is, on the possible *links* that the understanding can impose on the unrelated outcomes of empirical receptivity, in order to produce *a priori* structured (i.e. syntactical?) referential claims.

The synthetical approach we find in Kant’s *transcendental deduction* pivots on the non

¹For an extended discussion of Kant’s conception of logic with respect to ontology and grammar, see Capozzi (2006).

²A term Kant uses to denote the “unity of the act of collecting several representations under a common one”, (Kant (2010)).

empirical nature of logical syntax: logical judgements manifest indeed abstract forms of composition of concepts that we can employ in an infinite range of different occasions; these abstract conjunctive forms, which must be given *a priori*, are to be thought as the product of *synthesis functions* of the understanding: it is the operating of these functions which brings forth structured, analyzable (i.e. decomposable) representations. With Kant's own words,

But the conjunction (conjunctio) of a manifold in intuition never can be given us by the senses [...] To this act we shall give the general appellation of synthesis, thereby to indicate, at the same time, that we cannot represent anything as conjoined in the object without having previously conjoined it ourselves. Of all mental notions, that of conjunction is the only one which cannot be given through objects, but can be originated only by the subject itself, because it is an act of its purely spontaneous activity. The reader will easily enough perceive that the possibility of conjunction must be grounded in the very nature of this act, and that it must be equally valid for all conjunction, and that analysis, which appears to be its contrary, must, nevertheless, always presuppose it; for where the understanding has not previously conjoined, it cannot dissect or analyse, because only as conjoined by it, must that which is to be analysed have been given to our faculty of representation. Kant (2010)

In this sense, categories, the pure concepts which enable the *synthesis* of experience, force experience to fit into the net they produce, in order for its content to be *analyzed* as a composition (a “conjunction” indeed) of simpler representations: categories, in Kant's terminology, are the ultimate links to which every connection in thought can be reduced.

Kant extrapolated his list of categories from his table of logical judgements, that is, from the logical syntax he adopted in his courses of logic: categories thus embody the way

in which the logical tradition of his time formatted scientific language, so as propositional logic, predicate calculus etc. embody our contemporary way of formatting logic. The rules of our favorite sequent calculus, we could say, constitute thus the links through which we express our referential claims and which entitle their logical analysis.

The core of Kant's argument is that the transcendental step from *analysis* to *synthesis*, from decompositional forms to the ultimate conditions of composition, can unearth the "legal" background which sustains logical analysis: categories make logic appear in its authoritative, even "despotic" character. Understanding, following Kant, is in fact not in control of what categories demand for it, so that it can be deceived and deluded for not being able to exercise a critical power towards them: this was the main argument of transcendental dialectics, regarding the *transcendent* use of categories.

Understanding, at the mercy of categories, can be thus naturally driven towards empty directions, that is, to impose rules as if there were an object to which they could be referred, whereas just those rules can never constitute anything as an object for them; as Kant explains in transcendental dialectics,

For all these questions relate to an object, which can be given nowhere else than in thought. [...] If the conceptions in our minds do not assist us to some certain result in regard to these problems, we must not defend ourselves on the plea that the object itself remains hidden from and unknown to us. For no such thing or object can be given—it is not to be found out of the idea in our minds. We must seek the cause of our failure in our idea itself, which is an insoluble problem and in regard to which we obstinately assume that there exists a real object corresponding and adequate to it.

Kant (2010)

It appears then natural to ask whether a critical power can be exercised towards axioms, rules, notations, all the formal “bureaucracy” which administers our theoretical activity. Such a critical power, on the other hand, would seem to have no definite content from a Fregean, purely analytic, standpoint. Furthermore, Hilbert’s supreme court notoriously fails in its ambitions: indeed, Gödel’s second theorem directly puts into question the authority of the court: who can assure that the systems in which proofs of non-contradictions are formulated does not itself fall into contradiction?

Reformulating Kantian transcendental deduction of categories in our terms, we could then ask: by what right do our favorite formal systems play such a commanding role in logic? *Quid iuris?*³

2.3 *Mirror effects*

In this paragraph we discuss what appears to us as a *blind spot* of the method of analysis, namely the fact that, since syntactical notions are atomic, i.e. not analyzable, attempts at justifying systems of rules by means of semantical or (meta-)linguistic arguments often end up turning around in a circle.

For instance, the traditional explanation of the rule of *modus ponens* essentially makes use of an application of *modus ponens* (this was indeed the theme of Carroll (1895)): if you have a proof of A and a proof of $A \Rightarrow B$, then, since the latter means that, given A , you can conclude B , then you can conclude B , an argument which takes the explicit form of a *modus ponens* (similar cases can be constructed for most of traditional logical rules).

³Kant makes a celebrated distinction between a *quid facti* and a *quid iuris* question on the use of categories: the first regards the fact that categories effectively impose their form to experience, the second regarding indeed the right by which categories constitute its ultimate judge.

This form of circularity (called *pragmatic circularity* in Dummett (1991)) has the effect of highlighting the somehow authoritarian character of syntactic rules: one can in no way put them into question, since they repeatedly recreate themselves. As stressed in Dummett (1991), pragmatic circularity impedes any solution to a debate as to the acceptability of an inference rule:

If the justification is intended as suasive, then the pragmatic circularity will defeat its principal objective. That is to say, if the justification is addressed to someone who genuinely doubts whether the law is valid, and is intended to persuade him that it is, it will fail its purpose, since he will not accept the argument. Dummett (1991)

A typical response to this objection is given by the appeal to a semantical justification: logically correct rules are those which *preserve truth*. This means that one can provide a semantical theory following which, for any interpretation which assigns the value “true” to the premisses of the logically correct rule, the value of the conclusion in the same interpretation will be “true”: this is equivalent to the constraint that semantics must be *compositional*. The problem for this response is that the want of compositionality produces the result that semantical justification will closely reflect (though in “metalinguistic” disguise) the syntactical structure of what is to be justified: as an outstanding example, Tarskian semantics will take the formula $\forall x \exists y A(x, y)$ to be true in a model $\mathcal{M}$ exactly when *for all* $a \in M$ (where M , a set $\neq \emptyset$ is the support of $\mathcal{M}$) *there exists* $b \in M$ such that $A[a, b]$ is true in $\mathcal{M}$. Put in Kantian terms, it seems as if the subject had projected his motion into the movements of its object and then failed, as in Ptolemaic astronomy, to discern its trace.

If these forms of circularity aren't of any problem for purely semantical investigations

(as Tarski's ones were), it makes nonetheless semantic justification of no use when the acceptability of a system of rules is disputed. A remarkable example is to be found in Cellucci (2006), where Cellucci explicitly takes into consideration Dummett's pragmatic circularity: as he argues, if *modus ponens* is to be justified by an argument resting on an application of *modus ponens*, why shouldn't we justify *abduction*, that is, the logically incorrect inference

$$\frac{A \quad B \Rightarrow A}{B} (Abd) \quad (3)$$

by resting on an application of the rule (*Abd*)? In Cellucci (2006) two "abductive" justifications of abduction can indeed be found, the one based on semantics - i.e. truth-preservation -, the other styled "inferentialist", specifically proof-theoretical. If we stick to the semantic and to the inferentialist views, there would be thus no serious difference in quality between *modus ponens*, the most unassailable of deductive laws, and abduction, the typical example of a non deductive law.

Now, if there is no room for disputing the formatting imposed by formal languages, every rule being justified in the mirror, the radical consequence to draw would be that every claim of logical correctness isn't to be confronted than with itself, a strong form of relativism which seems philosophically very unsatisfactory.

In definitive, semantics and the linguistic description of rules do not seem to provide a satisfying answer (typically, a rule being explained by its metalinguistic *alter-ego*) to any demand for legitimization: so to say, in front of Kant's *quid iuris*, the machinery of analysis inexorably runs idle.

3 *The ambitions of transcendental syntax*

If we drop the criticisms of the preceding section, that is, if we drop the requirement that analytical tools, enhanced by rigorous presentations of language, be supported by a synthetical questioning of their form, and the objection that semantical justification can hardly transgress this form, we are left with a quite systematical and familiar frame (already depicted in the previous section).

On the other hand, in Girard (2011b) transcendental syntax is presented as a program aiming at finding a completely different foundation for logic:

La syntaxe transcendante substitue à la *sémantique* — i.e., à la métaphysique du «réel» — l'étude des *conditions de possibilité* du langage logique. Girard (2011b)

The reference to Kant is made explicit by the following remark:

[...] il est nécessaire de comprendre la part de nous-même dans les objets que nous étudions, d'où l'invocation de Kant — du moins de sa méthode — pour essayer de trouver, sinon un format définitif (qui serait alors nécessaire), du moins de sérier les problèmes pour ne pas objectiviser ce qui appartient au sujet ou subjectiviser le «monde». Girard (2011b)

The ambitions of the program are summarized by the following purpose: the development of a new format to represent logical proofs and types (quite in the lines of *geometry of interaction*) whose “normative” features (typing criteria, cut-elimination, completeness) are not to be searched for by reference to external structures, but find an explanation through geometrical criteria directly applicable to syntactical artifacts. This general point, which was sustained by a Kantian inspiration that we are trying to make explicit, has as

corollaries at least two other ambitions, that we discuss subsequently: the use of transcendental syntax tools to exert a critical power towards existing syntaxes, by putting together philosophical justifications and purely technical ones and the refinement of actual logical systems, along with a finer understanding of their inner constitutive principles.

Ambitious being the program, ambitious is also the name of the program: transcendental is, for Kant, a form of knowledge which is not directed to the objects of ordinary science, but to “the mode of our cognition of these objects, so far as this mode of cognition is possible a priori” (Kant (2010)); the issue of transcendental enquiry is therefore to identify the place of the subject in the universe of his multifarious knowledges, a place which makes it possible for those knowledges to manifest an objective relation with the subject himself.

Kant aimed at justifying the objective claims of mathematicians and physicians of his time, and at finding means to charge the metaphysical debate, so devoid of the clarity of the formers, of being a collection of “dialectical illusions”; nevertheless, he wanted these results to be the outcome of an investigation on the respective forms of argument and judgement of the mathematician, the physician and the metaphysician; it was, after all, an enquiry entirely directed to the categorial (i.e. syntactical?) criteria constitutive of the debates characterizing the disciplines. Never, in his arguments, Kant mentions the objects of those disciplines: what he disputes is whether they can ever have a definite object.

It is more or less in this spirit that the attempt at unearthing the constraints hidden in the rules of logic aims at an explanation of how syntactical constraints are related to traditional logical results such as general cut-elimination theorems, completeness and incompleteness theorems; it is again in this spirit that enquiries within transcendental syntax

should to be performed without reference to those entities whose representability crucially relies on the use of what is charge of being justified; this constraint was indeed the one making a difference, for Kant, between the transcendental enterprise and metaphysics.

For instance, the geometrical criterion of proof-nets provides an apparently internal explanation of the constraints imposed by logic typability (i.e. sequent calculus): it seems indeed to show that introduction and elimination rules do not fall from the sky, but are just those which prevent disputes to fall into (vicious) circles.

As a concrete example of how the criterion works (see Girard (2011b)), we take proof-nets for first order quantifiers (see Girard (1991)): this part of proof-net theory is the result of a geometrical reinterpretation of a genial trick adopted by Herbrand in the theorem which bears his name (see Herbrand (1930)), and which produces an apparently non circular explanation of quantifiers (on the contrary, as we mentioned above, the Tarskian explanation of quantified formulas like $\forall x \exists y A(x, y)$ is given in terms of quantifiers applied to the elements of the support of the models interpreting the formula). Herbrand's trick uses the technique of *unification* (see Herbrand (1930)), an algorithm for the solution of systems of equations between functional terms. To formulas of the form $\exists x \forall y B(x, y)$ Herbrand associates the equation

$$x = f(y) \tag{4}$$

which implicitly corresponds to the “Herbrandization” of the formula, i.e. its transformation into the formula $\forall f \exists x A(x, f(x))$ (the latter being valid if and only if the former is).

Now, whereas the semantic refutation of the invalid formula $\forall x \exists y A(x, y) \Rightarrow \exists y \forall x A(x, y)$ still makes use of quantifiers, Herbrand's refutation does not: it is indeed obtained by

considering the quantifier-free “Herbrandized” formula $A(g(y), y) \Rightarrow A(x, f(x))$ and then showing that the induced system of equations

$$\begin{aligned} x &= g(y) \\ y &= f(x) \end{aligned} \tag{5}$$

has no solution. The translation in terms of proof-nets is the following: one considers an arbitrary proof-structure⁴ representing an alleged proof of the formula; to any variable which is eigenvariable of a “forall” quantifiers a specific point of the net is associated; the system (5) is thus translated into the existence of a path in the net from the point associated to y to the one associated to x so as a path from the point associated to x to the one associated to y , i.e. a (vicious) cycle!

On the other hand, one easily sees that no equations are produced in the case of the logically valid formula $\exists x \forall y A(x, y) \Rightarrow \forall y \exists x A(x, y)$, to which a proof-net with no cycles can be naturally associated.

In addition to constitute a tool to understand logic from the mathematical structure of its languages, transcendental syntax has also been proposed as a tool to discuss and criticize ill-behaving syntaxes; this seems to us a possible way to reconcile the apparent gap between two rather antipodal approaches to logic: on one side logicians coming from a philosophical tradition (especially analytic), focused on semantical justification, and thus involved in the construction of logics enjoying a philosophically convincing semantical explanation, while being very poorly interested in the inner quality of their logical syn-

⁴Proof-structures are the equivalent in proof-net theory of λ -terms *à la Curry*, that is pure terms which do not necessarily correspond to correct proof-terms. Only those proof-structures which satisfy the correctness criterion (which have no vicious cycles) are called proof-nets and are sequentializable.

taxes; on the other side, logicians coming from a computer-science experience, poorly interested in philosophical explanations but very careful to the computational properties of their logical languages, crucial for programming purposes.

To make an example of a possible bridge between the two views, a fundamental thesis of transcendental syntax is that compositionality, both in the proof-theoretic (i.e. the fact that the provability of a compound formula is defined in terms of the provability of its components) and semantic (i.e. the fact that the truth of a compound formula is defined in terms of the truth of its component ones) sense, is grounded in cut-elimination; for example, that the provability of $A \Rightarrow B$ is function of the provability of A and B derives from the fact that, following the typing *à la Curry*, a (canonical) proof Π of $A \Rightarrow B$ is anything which “sends” a (canonical) proof Λ of A into a (canonical) proof Λ' of B and here the sense of this “sends” is made rigorous by saying that the *application* of Π to Λ *reduces* (through cut-elimination) to Λ' .

On the other hand, for systems not enjoying a convincing form of cut-elimination semantics can be compositional only *at the level of formulas*, but it will fail to reflect compositionality *at the level of proofs*, i.e. to interpret proof composition. For instance, modal logic *S5*, admits of a compositional semantics (Kripke models) for which, since its standard syntax does not enjoy cut-elimination, one can find an example of proof composition which is not reflected at the semantical level: take the following cut, for which no direct procedure of elimination is possible

$$\frac{\frac{\vdots}{A \vdash \Diamond A} \quad \frac{\frac{\vdots}{\Diamond A \vdash \Diamond A}}{\Diamond A \vdash \Box \Diamond A}}{A \vdash \Box \Diamond A} \text{ cut} \quad (6)$$

and which is represented in Kripke models as a perfectly well-behaved composition (through an opportune identification of variables standing for possible worlds)⁵.

Finally, the focus on dynamics may allow to discover “subjective” redundancies in known formalisms: bureaucratic, as one might say, aspects of syntax are those that are irrelevant to the behaviour of proofs (i.e. their interaction through cut–elimination) that is, i.e. to its use (or its meaning, following Wittgenstein’s notorious equation - Wittgenstein (2009), §43 - ?). Proof-nets, for instance, actually perform a quotient on sequent calculus proofs, since different proofs can be associated to distinct sequentialization of the same net, thus erasing about irrelevant information.

It must be said that, whereas for a restricted class of proof-nets (relative to *multiplicative linear logic*) this quotient has a clear and convincing status, in the general case (concerning *additive* and *exponential* linear logic) it is still no clear where to draw the line between “essential” and “inessential” sequential information; this problem, which constitutes one of the main line of technical research in transcendental syntax, comprising the crucial theme of computational complexity (see Girard (2012)), really appears like (and is presented by Girard as) a sort of “transcendental” enquiry on the positioning of the subject and his “epicycles” (the “inessential” components of syntax) with respect to logic.

Here is how Girard describes the sense of working on transcendental syntax:

If we exclude divine revelations, the only possibility consists in making things interact with alter egos. In the case of pasta, one alters the recipe and sees whether it tastes the same. Typically, put the salt before boiling, you will notice no difference; push the cooking time to 15mn and you get glue. To sum up, restrictions are not out of a

⁵Needless to say that there are variants of *S5* which enjoy cut-elimination; the example is intended only to evidence the connections between the two forms of compositionality.

Holy Book, but out of use. And use is internal, i.e., homogeneous to the object. Girard

(2003)

The challenges discussed so far seem to converge on what we can call, freely taking inspiration from Kant, a *criticist stand on syntax*: by this we mean a philosophical position, still to be delineated in detail, which on the one side refuses the “mirror effects” of semantical explanation, as we have exhaustively explained, and on the other does not resign to the conventionalist refusal of any explanation in logic.

After all, the conclusions of the arguments of the preceding section, namely that logical analysis, as conducted within a given formal system, presupposes a synthetic explanation of the syntactical tools involved, and that these forms of explanation should not end up into a semantic “tail-chasing” (rules justified by “meta- rules”), constitute the main philosophical tenets of the criticist position.

However, since we believe that transcendental syntax, as a technical, though philosophically oriented, program, must be evaluated on the field of its technical perspicuity, only an honest assessment of its results (those already obtained and, most of all, those yet to come) will tell for the validity of this proposal.

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